

Hardik Gupta

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PROFESSIONAL SUMMARY

Electronics & Communication Engineering undergraduate with hands-on experience in hardware component lifecycles, manufacturing operations, engineering change management, and technical project management. Skilled in **Engineering Change Notices (ECN)**, **Product Change Notifications (PCN)**, **Config/Order Bill of Materials (BOM)** verification, **Stage-Gate milestone governance**, and **Excess & Obsolete (E&O) inventory mitigation**. Adept at bridging technical specifications with operational deliverables and driving cross-functional alignment across R&D, Quality, Sourcing, and Plant Operations.

EDUCATION

PES University, *B.Tech in Electronics & Communication Engineering* Aug 2023 – Present — Bengaluru, India
– **Relevant Coursework:** Microcontrollers, System Architecture, Embedded Systems, Signals & Systems, Data Analytics

Lotus Valley International School, *Class 12th (CBSE): 86% | Class 10th (CBSE): 93%* Apr 2020 – Mar 2023 — Noida, India

TECHNICAL & MANAGEMENT SKILLS

Product Change & Ops: Engineering Change Notices (ECN), Manufacturing Change Notices (MCN), Product Change Notifications (PCN), Config & Order BOM Audit, Process of Record (POR), Excess & Obsolete (E&O) Inventory, Six Sigma / CIP, Obsolescence Mgmt
Project & Stakeholder Mgmt: Stage-Gate Milestone Governance, End-to-End Product Lifecycle Mgmt (PLM), RACI Stakeholder Alignment, Risk Mitigation, Cross-Functional Execution (R&D, QA, Sourcing, Ops)
Data Analytics & Viz: Data Variance Reconciliation, KPI Metrics Reporting, Power BI, **Advanced MS Excel** (Pivot Tables, VLOOKUP, Data Modeling), SQL (Basics), Python (Pandas, NumPy)
Tools & Platforms: MS Office Suite (**Excel**, Word, PowerPoint, SharePoint), SAP / ERP Systems (Awareness), Jira, Git, GitHub, VS Code, Google Colab, LaTeX

EXPERIENCE

Bharat Space Education Research Centre (BSERC) Dec 2025 – Jan 2026
Systems Engineering Winter Intern New Delhi, India
– **Evaluated aircraft design change requirements** using MATLAB and XFLR5, modeling stability parameters and delivering data reconciliation reports to cross-functional engineering stakeholders.

ColorHome Lighting Jul 2025 – Aug 2025
Operations & Product Change Summer Intern Noida, India
– **Managed ECN and MCN implementation workflows** for 9W LED bulb assemblies, auditing component sourcing, **Config BOM and Order BOM** structures, and production dispatch effectivity dates.
– **Executed supply and demand inventory assessments**, collaborating with Supply Chain to evaluate material rework, minimize **Excess & Obsolete (E&O) inventory**, and optimize production milestones.
– **Interfaced cross-functionally** across Design, Sourcing, and Quality Assurance teams to document Process of Record (POR) standards and maintain SMT line quality compliance.

TECHNICAL & MANAGEMENT PROJECTS

Vocab Ledger – Mobile Product Lifecycle & Store Deployment ► Jul 2026
– **Directed end-to-end product lifecycle** and component change tracking for a native Android app (Kotlin, Compose) with Google Gemini AI API integration; published live on **Google Play**.
– **Spearheaded binary optimization** using custom R8/ProGuard minification rules, achieving a **44% reduction** in app binary footprint (from 19.7 MB to 11.0 MB).

Automated PCB Defect Detection & Quality Assurance System 📷 Apr 2026
– **Engineered an automated quality inspection pipeline** using YOLOv5s/v8s on 693 PCB images, classifying 6 defect types with **97.4% mAP@0.5** and **95.2% recall**.
– **Designed a 4-tier severity matrix** evaluating defect area and position to deliver automated **PASS/FAIL verdicts** and repair recommendations, driving **Continuous Improvement (CIP)**.

Industrial Defect Inspection & Batch Processing Pipeline 📷 Nov 2025
– **Built an automated OpenCV inspection pipeline** using local variance analysis and morphological filtering to isolate irregularities across batch manufacturing datasets.
– **Engineered a batch-processing system** to calculate quality KPI metrics, annotate defect locations, and generate automated inspection compliance reports.

CERTIFICATIONS

AI, ML & Data Analytics with Drone Tech – IIT Hyderabad (TiHAN) Jun 2025 – Feb 2026
– Hands-on training in machine learning models, **data analytics**, hardware building blocks, sensor integration, and system architecture from **IIT Hyderabad**.

Microsoft Office Specialist: Excel 2013 – Microsoft

– Certified in spreadsheet data management, formulas, pivot tables, and data modeling in MS Excel.